Adapting the MINUTE-ChIP Pipeline for High-Throughput MBD-seq: Cross-Platform Validation Against Illumina MethylationEPIC 850k Array

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Abstract

This study presents the implementation and validation of a novel epigenetic profiling workflow, MINUTE-ChIP, for the characterization of the DNA methylome in human peripheral blood mononuclear cells (PBMCs). Despite the importance of DNA methylation as a clinical biomarker, high-throughput sequencing methods often suffer from high costs and technical variability. This project addresses these challenges by documenting the MINUTE pipeline for NGS-based methylation capture and comparing its performance against the Illumina MethylationEPIC (850k) array. Following quality control, three distinct enrichment fractions (In-house MBD, Commercial MBD, and Unmethylated CXXC) were analysed.

Principal Component Analysis (PCA) and fingerprinting revealed that the In-house MBD method achieved better separation from input control compared to the commercial kit, which exhibited signs of higher background noise-to-signal. Genomic distribution analysis showed that while the commercial kit captured broader distal intergenic regions, the In-house method demonstrated higher specificity for proximal promoter regions (≤ 1kb). Biological validation confirmed cell identity, with active PBMC markers (e.g., CD45, CD4, CD8) found in unmethylated fraction and pluripotency factors (e.g., NANOG) correctly identified as methylated. Cross-platform comparison identified a high-confidence "core" set of 6,888 methylated genes common to all methods. While the 850k array remains the gold standard for absolute quantification, the MINUTE-ChIP pipeline provides a scalable, cost-effective, and unbiased alternative for genome-wide discovery in precision medicine.

Keywords: MINUTE-ChIP, MBD-seq, Illumina MethylationEPIC 850k, Next-Generation Sequencing (NGS), Affinity-based Enrichment, DNA Methylation, CpG Islands, Transcription Start Site (TSS), Epigenetics, PBMCs, Precision Medicine, Bioinformatics, deepTools, Bedtools, R.

Introduction

Epigenetics and Precision Medicine

Epigenetics describes how the genetic code is regulated, allowing cells to differentiate and adapt to various environments without changing the DNA sequence. Genes can either be silenced or activated, which determines cell fate and differentiation. This has been seen in response to infections (Zhang & Cao, 2019), chronic diseases (Fischer, 2020), cancer (Yu et al., 2024) and outer stimuli (Turner, 2009). Regulation can be on chromatin level, where histones can be modified with methyl ($-CH_3$), acetyl ($-COCH_3$), phosphate ($-PO_4$) groups or ubiquitin, changing how tightly the chromatin is packed; heterochromatin being 'closed' for transcription and inaccessible to transcription machinery and euchromatin being 'opened' for transcription. Another type of regulation "DNA methylation (DNAme)" is directly on the DNA sequence and works by adding a methyl group to C5 position of a cytosine base, most commonly followed by a guanine (CpG). Where the genome is rich in CpG sites is referred to as CpG islands and they are typically present in gene regulatory regions such as gene promoters. Both histone modification and DNA methylation are tightly linked and depended on each other (Cedar & Bergman, 2009).

Epigenetics is widely used in medical diagnostics because it gives plethora of information about how lifestyle and environment affect gene expression and additionally, epigenetic changes are reversible which is an important feature for development of novel therapies. In cancer one of the main hallmarks is epigenetic changes that support cancer growth by either silencing tumor suppressor genes (TSGs), referred to as TSG hypermethylation or activating oncogenes via promoter hypomethylation. Consequently, epigenetics has become central to precision medicine by utilizing epigenetic changes to detect diseases early and provide more accurate prognosis and treatment. An example of epigenetics usage in cancer diagnostics was shown by Qvick et. al. who demonstrated that DNA methylation pattern changes in circulating cell-free DNA (cfDNA) can be used to distinguish patients with various cancers from healthy controls (Qvick et al., 2025). Another example lies in disease prognosis, where using DNA methylation patterns in SARS-CoV-2 infection can reveal longstanding epigenetic changes as suggested by Huoman et.al. This could explain long term odour perception loss after COVID-19 (Huoman, J., 2022).

With sequencing technologies becoming gold standard for various experiments, the need for handling high-throughput sequencing data is increasing and with it developing robust pipelines for data analysis.

Bioinformatics for DNA methylation

Most common modern technologies for studying DNA methylation include bisulphite conversion-based methods (such as whole genome bisulphite sequencing (WGBS) and reduced representation bisulphite sequencing (RRBS)), affinity enrichment techniques (like methylated DNA immunoprecipitation sequencing (MeDIP-Seq), Methyl-CpG-Binding Domain sequencing (MBD-seq)) and microarray platforms (for instance Illumina Infinium arrays) (Mehrmohamadi et al., 2021). Besides bisulphite conversion, there are also enzymatic methods available such as Enzymatic Methyl-seq (EM-seq) by Illumina or probe panels like TWIST by Twist Bioscience. Key challenges in current DNA methylation workflows include high DNA input requirements, high costs, and significant technical variability. Nowadays, many research groups are shifting from WGBS to more sensitive and non-destructive methods, as the harsh chemical treatment from extreme pH and high temperatures make it difficult to analyse samples with limited input such as liquid biopsies (cfDNA) or rare cell types (Whalley et al., 2021).

Another challenge lies in the limitation of short read sequencing, which fails in accurate capturing of repetitive elements in the genome such as short tandem repeats (STRs), microsatellites, minisatellites, transposable elements, high GC-rich regions, and more (Cechova, 2020).

To manage batch variability, standard pipelines often rely on mathematical batch correction methods like Surrogate Variable Analysis (SVA), which leverages matrix decomposition techniques to estimate latent sources of variation (Leek et al., 2010). While SVA helps remove noise, it introduces significant downsides. First, it alters the raw data values, which can be problematic for downstream tools like *limma* and *DESeq2* or *edgeR* (used for RNAseq data) that utilize negative binomial distributions and require raw integer counts for accurate modelling (Love et al., 2014). Second, mathematical correction might obscure true biological signals, particularly when detecting rare biomarkers where only a small percentage of samples are positive.

To address these limitations, we propose adapting a quantitative multiplexing strategy. The Multiplexed Indexed Unique-molecule Tagmentation (MINUTE-ChIP) pipeline has successfully demonstrated this for histone modifications, offering reproducible, scalable analysis (Kumar et al., 2025). By running fewer experiments with more replicates (up to 12 samples analysed at once), the MINUTE pipeline physically reduces technical variability, thereby minimizing batch effects at the source.

This adaptation is feasible because we utilize Methyl-CpG Binding Domain (MBD) capture (MethylCollector™ MBD Capture Kit, Active Motif). Unlike bisulphite sequencing, which chemically converts C to T, MBD is an affinity enrichment method that pulls down methylated DNA similarly to

how an antibody pulls down histone-bound DNA in ChIP-seq. Consequently, the resulting data structure mirrors ChIP-seq, allowing us to repurpose the robust MINUTE architecture to standardize DNA methylation analysis and improve efficiency.

Different methods need different analysis approaches

The primary distinction between Illumina 850k Epic Array and Epigenica's MINUTE (NGS-based capture) lies in how methylation is translated into a readable signal.

Illumina 850k Array is a direct measurement platform that uses two probes, Type I and Type II, to bind specific CpG sites. It provides a ratio of methylation, commonly referred to as the β-value, where $\beta = 0$ means that the region is completely unmethylated, and where $\beta = 1$ means that the specific locus is 100% methylated. The output files are *.IDAT* which are binary files containing raw red/green (Type I/Type II) fluorescence intensities. Primary tools to analyse this type of data include most commonly R and packages *minfi, ChAMP, SeSAMe* designed for normalization of array data. Resolution of the method is single base at 850,000 predefined locations. Overall, this means that the array provides higher quantitative sensitivity for hypo- and hyper-methylated sites but is limited to the fixed probes on the chip.

Epigenica's approach uses NGS following an enrichment step, like MBD-seq. It provides a relative signal intensity measured in Reads per Millions (RPM) where a higher density of reads in a genomic peak or domain indicates higher regional methylation. Primary tools to analyse this type of data include utilizing Bowtie2 for alignment and MACS for peak calling. The MINUTE pipeline designed for preprocessing of this data generates output such as *.fastq, .bam, .BigWig* files which are sequencing reads aligned to the genome and converted to signal tracks. Resolution is limited to fragment size (100-150bp) and capture efficiency of the MBD. All in all, this capture method gives broader regional coverage (peaks/domains) and can detect methylation in areas not covered by array probes but lacks the single base precision of bisulphite-based methods.

The Objective of the Project

This project aims to re-engineer the MINUTE pipeline to support DNA methylation analysis. Both MBD-seq and ChIP-seq require mapping to a reference genome and identifying enriched peaks, which means that after adjusting parameters for CpG density, peak-calling modules can be applied to DNAme. The data generated according to the Epigenica™ protocol is compared to the Illumina MethylationEPIC 850K array (Pidsley et al., 2016). Finally, the biological conclusion will be compared between the two methods by answering whether both methods identified the same genes as 'Methylated' and 'Unmethylated'.

Materials and methods

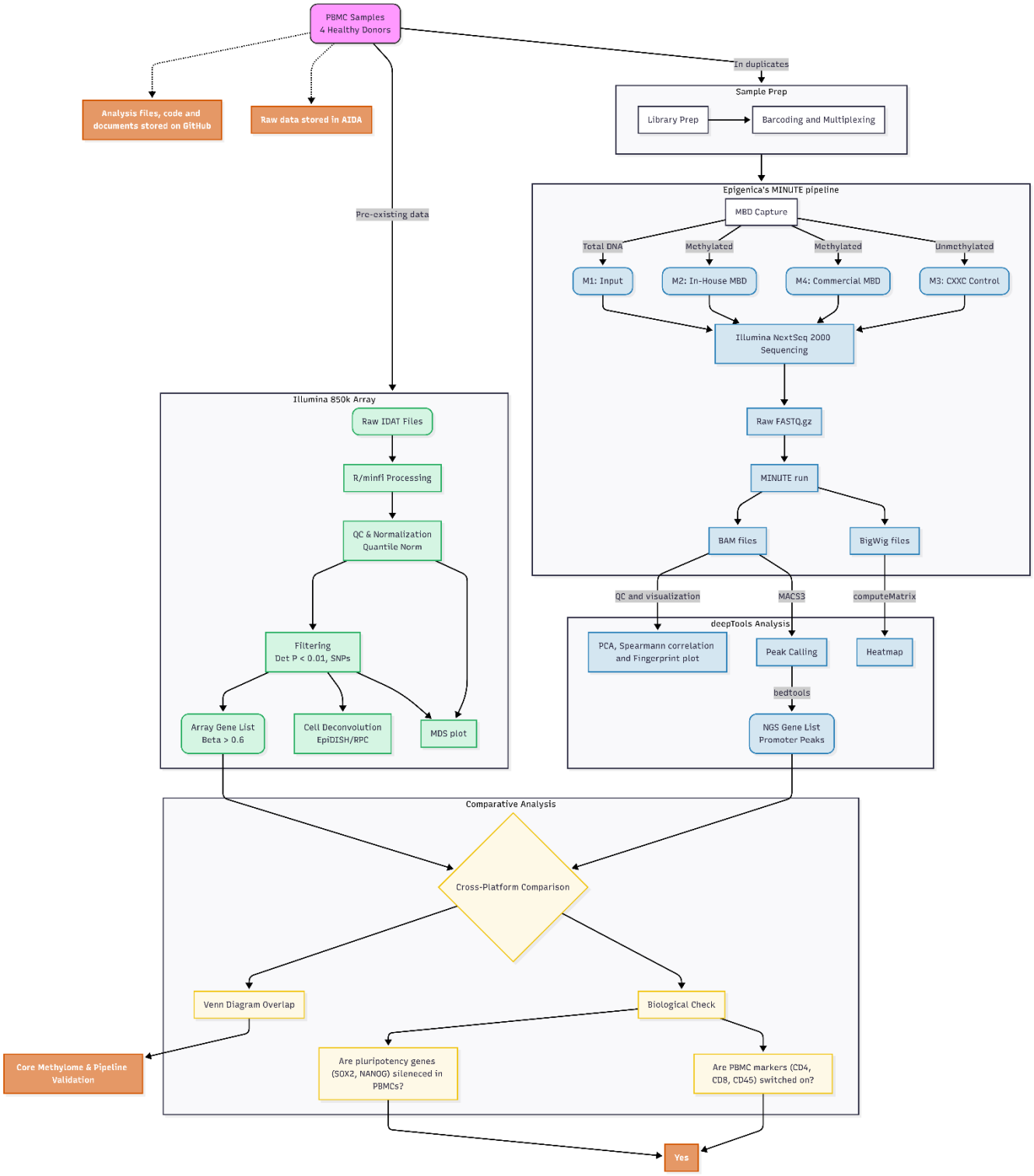


Figure 1. Overview of the project showing data processing and analysis using the MINUTE pipeline for NGS and R for the Illumina Array.

Input samples and library preparation

Peripheral blood mononuclear cells (PBMCs) from four healthy male donors were utilized, which were previously analysed and documented by Huoman et al. (2020). The same four donor samples, preserved as aliquots at -80°C, were used to evaluate the Epigenica protocol (Kumar et al., 2025). Libraries were prepared using the EpiFinder™ GenomePro Kit (1 million PBMCs per sample). The standard protocol was modified to bypass histone immunoprecipitation in favour of targeted methylated DNA enrichment. All samples underwent standardized processing, including lysis, enzymatic fragmentation, end-repair, and adapter ligation.

Illumina 850k array data analysis

a. Array Processing

Raw *IDAT* files were processed using the R package *minfi* (v4.3.3). The workflow included Quantile Normalization to facilitate comparison between donors.

b. Probe Filtering

Poor-quality probes were identified via detection p-values (>0.01), and probes associated with sex chromosomes, SNPs, or cross-reactivity were excluded to refine the dataset.

c. Gene List Generation

To enable a direct biological comparison between the microarray and NGS platforms, a high confidence "Methylated Gene List" was derived from the filtered array data using a multi-step bioinformatic approach. Analysis was restricted to probes located within gene regulatory regions, specifically the TSS1500 (Transcription Start Sites, 1500 bp upstream of the TSS), TSS200 (200 bp upstream), and the 5'UTR. Beta-values were averaged across all four donors to identify methylation patterns characteristic of the PBMC lineage, rather than donor-specific variation. Only probes with an average beta-value greater than 0.6 were selected to ensure that the genes identified are truly silenced and likely to produce a strong enrichment signal in MBD-seq. Probe IDs were mapped to their corresponding UCSC RefGene names using the *IlluminaHumanMethylationEPICanno.ilm10b4.hg19 package*. Because single genes were often covered by multiple probes, the final list was collapsed to unique gene symbols to prevent redundancy.

CpG Capturing and Enrichment

For each of the four donors, four conditions were established to evaluate the performance of different capture methods in duplicate:

- **M1 (Input):** Total DNA without selection, serving as a baseline control.
- **M2 (In-house MBD)**: Enrichment for methylated CpG sites using proprietary reagents.
- **M3 (CXXC):** A control capturing unmethylated CpG sites to validate the mirror-image distribution of methylated signal.
- **M4 (Commercial MBD):** Enrichment for methylated CpG sites using a commercial MethylCollector™ kit for benchmark comparison.

Sequencing

Libraries were sequenced on an Illumina NextSeq 2000 platform using a 100-cycle P3 reagent kit, NextSeq™ 2000 P3 Reagent Cartridge (XLEAP-SBS™ P3 100 cycles) (ref. 20101836, Illumina) and NextSeq™ 2000 XLEAP-SBS™ P3 Flow Cell (ref. 20094372, Illumina).

Epigenica data analysis: Bioinformatics Pipeline

Data analysis was conducted on the AIDA Data Science Platform using a 64-bit Linux environment. The MINUTE pipeline was implemented following the multiplexed chromatin immunoprecipitation protocols established by Kumar et al. (2025).

- **Pipeline Setup:** Conda and Bioconda environments were utilized to manage dependencies. The human genome (hg19) was indexed using *Bowtie2*, with the exclusion of specific problematic genomic regions defined in the BED exclude files. The exclusion list was derived from the ENCODE Project's Unified Blacklist for the hg19 reference genome and consisted of telomeres, centromeres, and highly repetitive satellite DNA that typically exhibit high background signal levels. Excluding those ensured that the identified peaks with MACS3 represent true enrichment and not non-specific reads.
- **Processing:** Configuration files including *barcodes.tsv, minute.yaml, groups.tsv*, and *libraries.tsv* were edited according to the experiment to facilitate the automated minute run.
- **Outputs:** The pipeline generated a 'final' directory containing demultiplexed *FASTQ* files, *BAM* alignments, and *BigWig* coverage files, alongside a 'reports' directory for MultiQC assessment.
- **Quality control report:** Quality control for the sequencing data was consolidated into a comprehensive MultiQC report to evaluate the technical success of the library preparation and

sequencing run (link below in 'Code availability'). Key figures were selected from the report including barcode representation, global reads scaling, GC content distribution, insert size profiles, distribution of reads across individual chromosomes and sequence quality assessment via Phred scores (Figures 16-21). The figures can be found under supplementary figures in the Appendix.

Data Visualization and Quality Control

The *deepTools* command lines were employed for secondary analysis.

a. Normalization and Scaling

Read counts were calculated across genomic bins using *multiBamSummary*. To ensure accurate visualization, Principal Component Analysis (PCA) was performed via *plotPCA*.

b. Metagene Analysis

To visualize signal distribution, heatmaps and summary profile plots were generated using the commands *computeMatrix* and *plotHeatmap*. These plots centred on Transcription Start Sites (TSS) and Transcription End Sites (TES) to assess the enrichment of signal across gene bodies.

c. Peak Calling

Regional enrichment of methylated sites was identified using *MACS3*.

d. Annotation

Identified peaks were intersected with the GENCODE hg19 gene map using *Bedtools* to generate specific methylated gene lists for each capture method.

Overlap Analysis - Gene-Specific Verification and Cell Identity Assessment

Data generated from the MINUTE-ChIP pipeline were compared against existing Illumina MethylationEPIC (850k) array data. The final comparison of gene lists was conducted using Venn diagram analysis to determine the concordance between the In-House MBD, Commercial MBD, and Array platforms. To verify the biological accuracy of the methylation capture, a targeted gene-identity analysis was performed. This was achieved through the following technical steps: (i) Gene list generation, (ii) Method-specific sorting, (iii) Identification of overlapping genes, (iv) Marker searching, (v)'Reality check' of the biology.

In detail: Gene lists were generated by intersecting MACS3-identified peaks with the GENCODE hg19 reference map using *bedtools intersect*. Lists were categorized into "Active/Unmethylated" (derived from M3 CXXC capture) and "Silenced/Methylated" (derived from M2/M4 MBD capture). For

the array, probes with β-values > 0.6 were classified as methylated and extracted as a list. The command-line utility *comm* was used to identify overlapping genes between the In-House MBD, Commercial MBD, and 850k Array gene lists to isolate a "Core" methylated gene set. Specific cell-identity markers were queried within these text-based gene lists using the *grep* command to confirm their epigenetic status. Known PBMC lineage markers (e.g., *CD45, CD4, CD8*) were verified in the unmethylated (M3) list to confirm active cell identity, while pluripotency factors (e.g., *POU5F1*, *NANOG*) were searched in the methylated (M2/M4) lists to confirm proper epigenetic silencing of hematopoietic genes.

Code availability

Link to GitHub https://github.com/Lerm-Lab/Bioinformatics_research_project

Data availability

The datasets presented in this study are not readily available because of ethical approval. Requests to access the datasets should be directed to the PI of the study Professor Maria Lerm, maria.lerm@liu.se.

Data storage

Sensitive patient data (raw IDAT, FASTQ.gz, BAM, BigWig files) is stored in the Data Science Platform AIDA. Data analysis files, pipelines, code and other documents are stored in the private directory on GitHub of Lerm-Lab.

Results

Section 1: Illumina 850k Epic Microarray Analysis

Analysis of the Illumina 850k data confirmed high sample quality (Figure 2) with mean detection p-values well below the 0.01 threshold (Figure 3). Cell type deconvolution using the EpiDISH/RPC method (Figure 6) revealed a typical PBMC profile for all donors, dominated by CD4+ T-cells, CD8+ T-cells, and Monocytes (Figure 6). The high consistency in cell type proportions between donors ensures that downstream methylation differences are biological and not due to shifts in leukocyte populations.

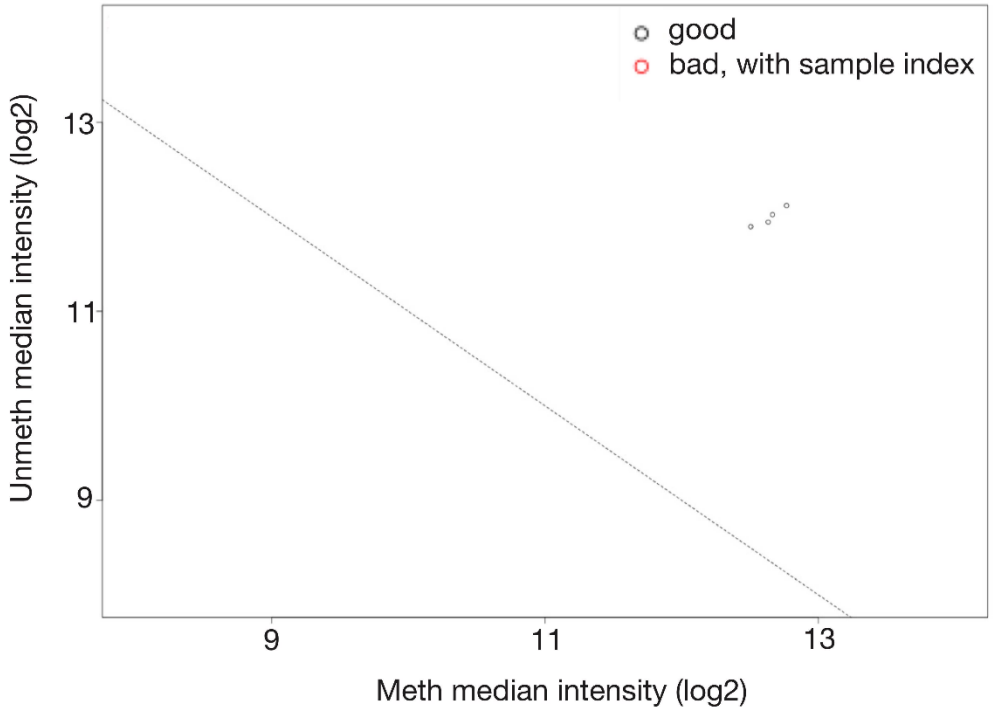


Figure 2. Quality control assessment based on median methylated and unmethylated signal intensities. The scatter plot illustrates the log2 median methylated versus log2 median unmethylated signal intensities for each of the four donor samples. Samples located in the upper right quadrant (labeled "good") signify high-intensity signals in both channels, indicating successful hybridization and sample processing.

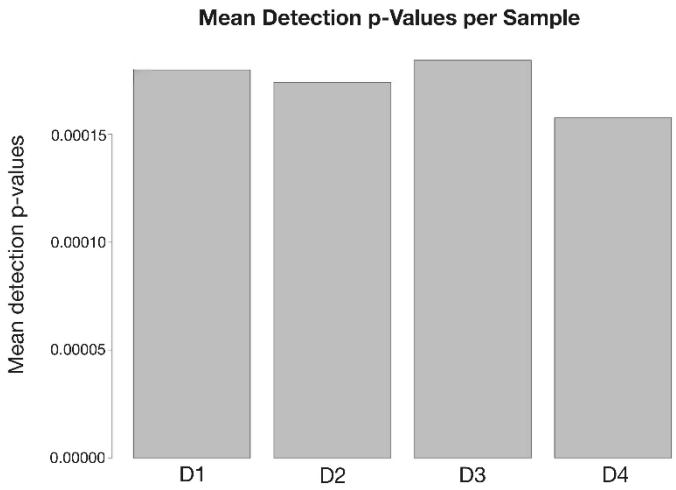


Figure 3. Mean detection p-values per sample. Plotted using *detectionP* function in R, which calculates p-values by comparing the total signal (Methylated + Unmethylated) of each probe to the background signal level which is estimated from the negative control probes.

Beta value distribution before and after normalization (Figure 4) showed stratified quantile normalization, which reduces inter-sample technical variation and is used when global differences between samples are not expected (e.g. in one-tissue datasets such as blood). Furthermore, the methylation data (beta values and M-values) was visualized using MDS plots (Figure 5).

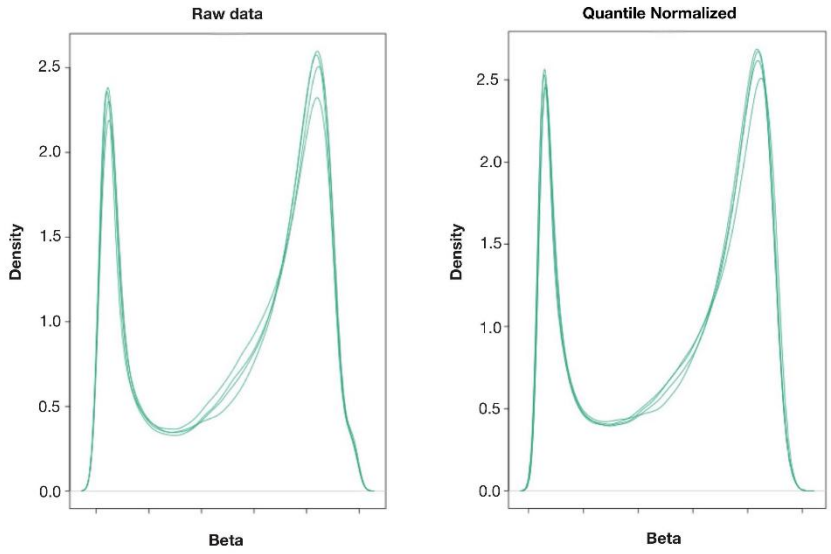


Figure 4. Distribution of beta-values before and after normalization. Density plots visualizing the frequency of beta-values for the four PBMC donor samples. Raw data (left): The initial distribution showing technical variation between donor samples. Quantile Normalized (right).

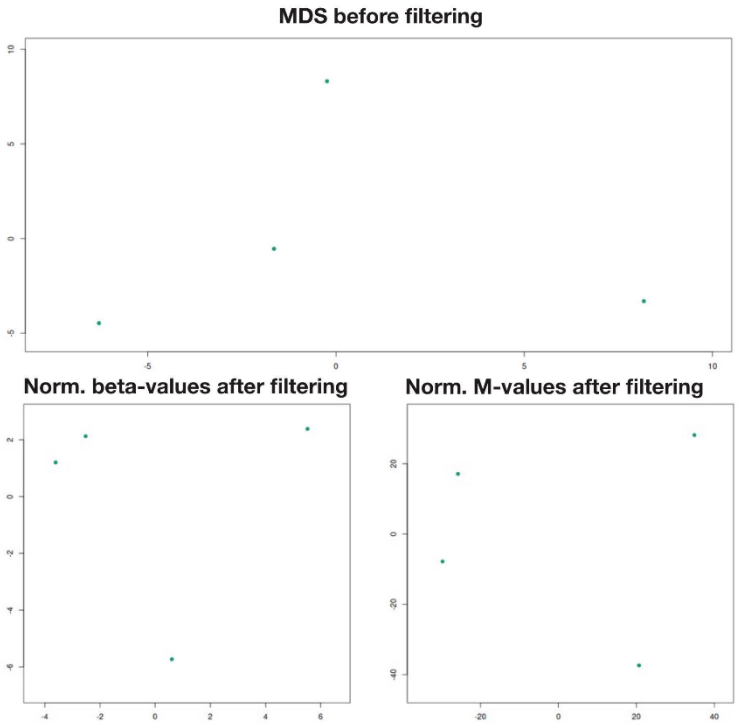


Figure 5. Multidimensional scaling (MDS) analysis of Illumina 850k data. Collective assessment of data quality and sample relationship following processing with *minfi* package. (Top) MDS plot of raw data prior to filtering. (Bottom left) MDS plot generated using normalized beta-values after filtering for poor-quality probes, SNPs, and cross-reactive sequences. (Bottom right) MDS plot based on normalized M-values (Mvalue = log2(M/U) after identical filtering steps.

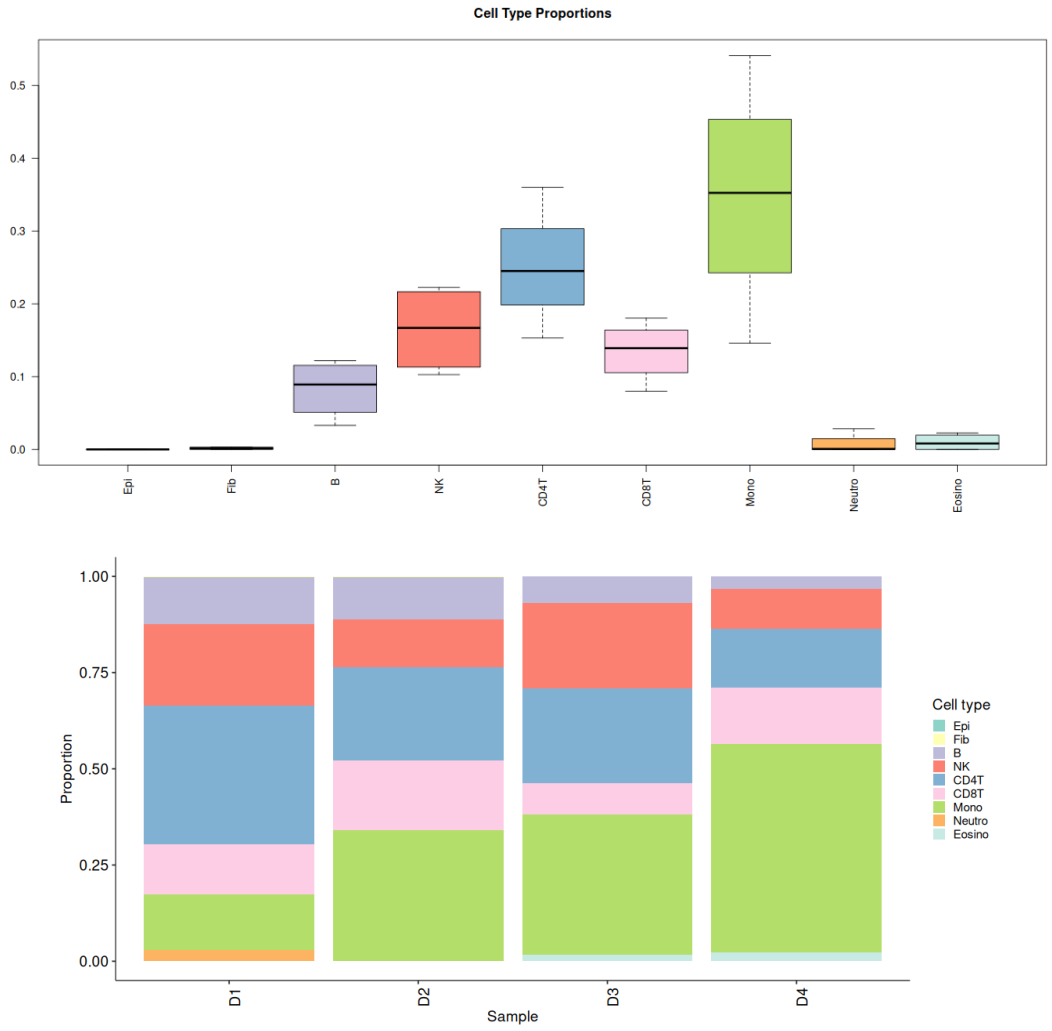


Figure 6. Estimated cell type composition and proportions across donors. Visualization of leukocyte populations derived from DNA methylation data using the *EpiDISH* package in R. (Top) Boxplot summarizing cell type proportions across the cohort. Analysis identifies monocytes and CD4+ T-cells as the predominant populations, consistent with typical PBMC compositions. (Bottom) Stacked bar plot illustrating the relative distribution of cell types for each individual donor.

Section 2: MINUTE Enrichment and Global Profiling

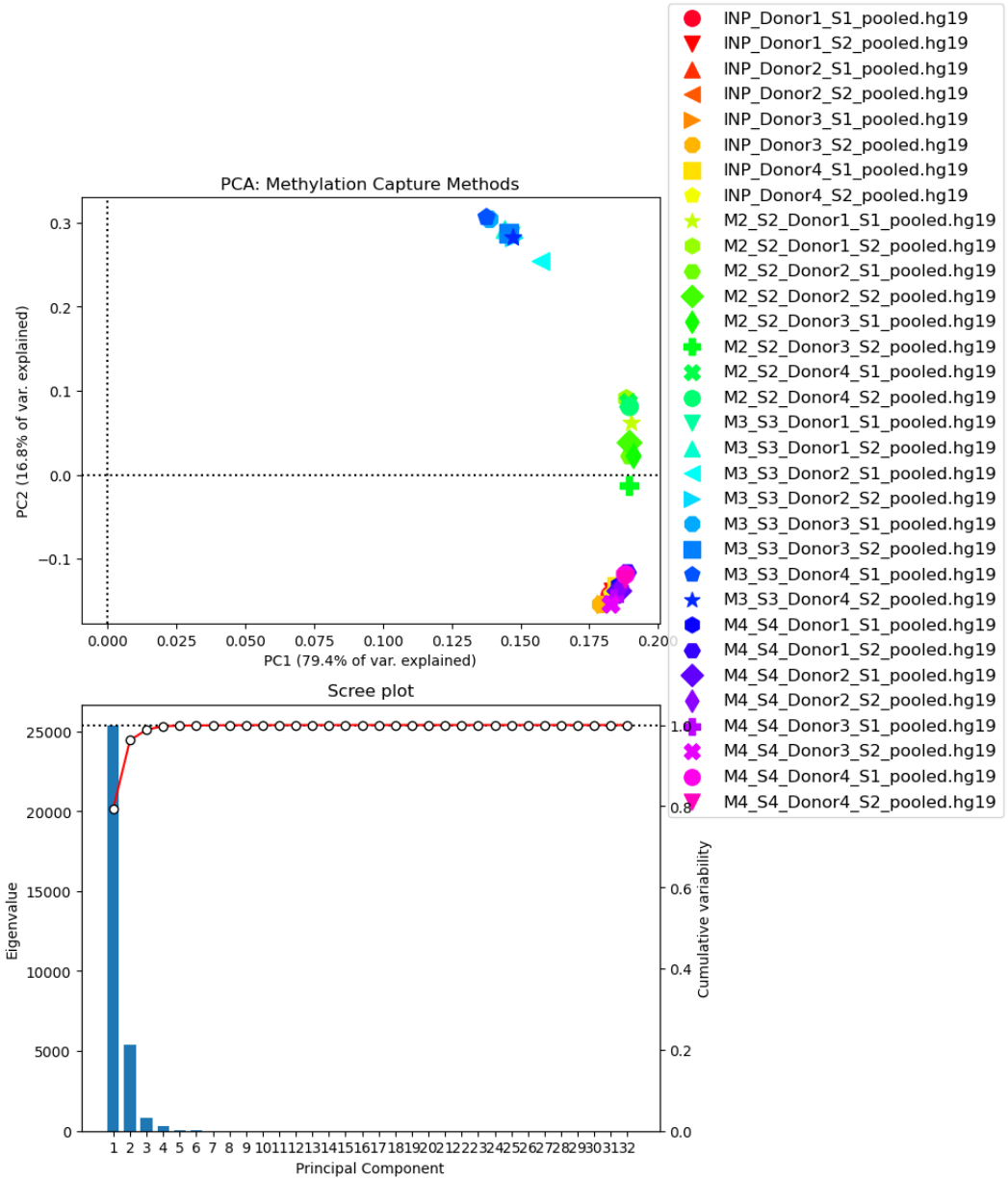


Figure 7. PCA plot (top) and Scree plot (bottom) of methylation capture methods. Scree plot confirmed that most of the variation is captured within PC1 and PC2.

Data Quality and PCA

PCA revealed that PC1 and PC2 explained 79.4% and 16.8% of the total variance, respectively (Figure 7) and demonstrated clustering based on the capture method in M2 and M3 samples. The M2 fraction ('In-house' MBD) shows the greatest separation from the Input, indicating successful enrichment of methylated regions. The M4 fraction (High Stringency) clusters closer to the Input, likely due to lower DNA yield in hyper-methylated regions, resulting in a higher background-to-signal ratio. Replicates within each method clustered tightly, demonstrating high technical reproducibility, while variation along PC2 likely reflects secondary sample-specific effects (Figure 7).

Spearman correlation heatmaps (Figure 8) confirmed high reproducibility between replicates, while effectively separating methylated (M2, M4) from unmethylated (M3) and Input fractions.

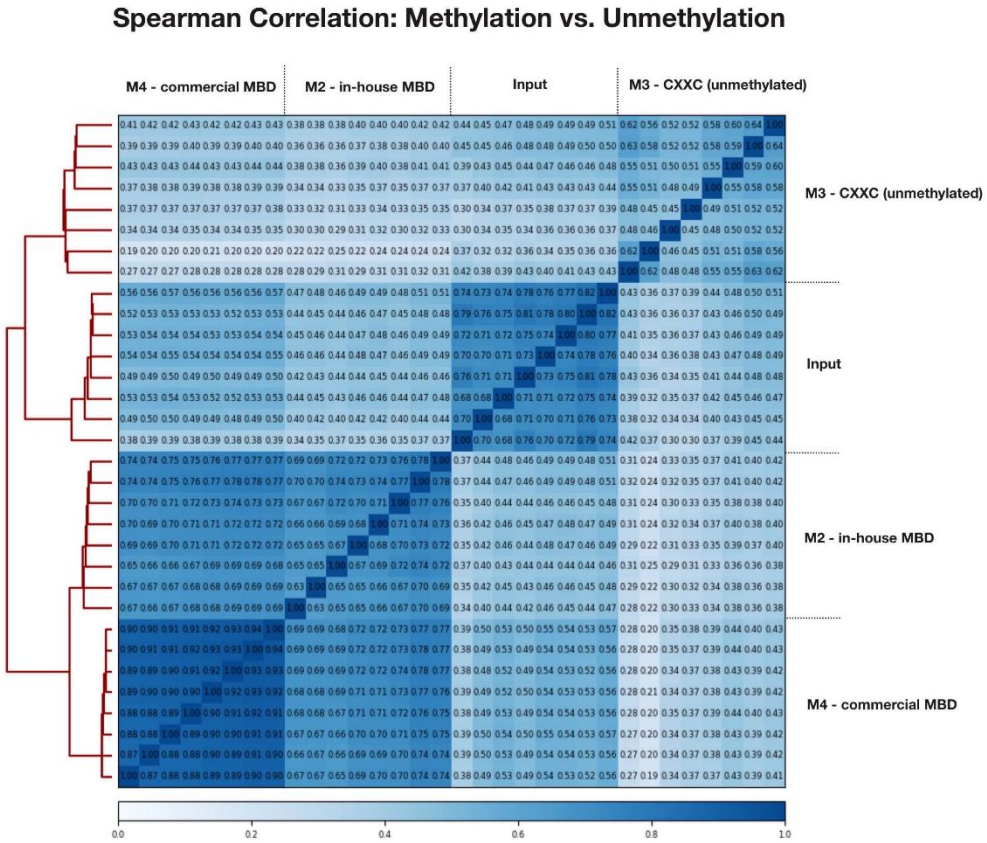


Figure 8. Spearman Correlation plot between all the replicates in all donor samples. Replicates correlate more to one another, rather than to other methods, which validates the method reproducibility.

Fingerprint plot (Figure 9) indicated significant enrichment in MBD-captured samples compared to the background of the Input.

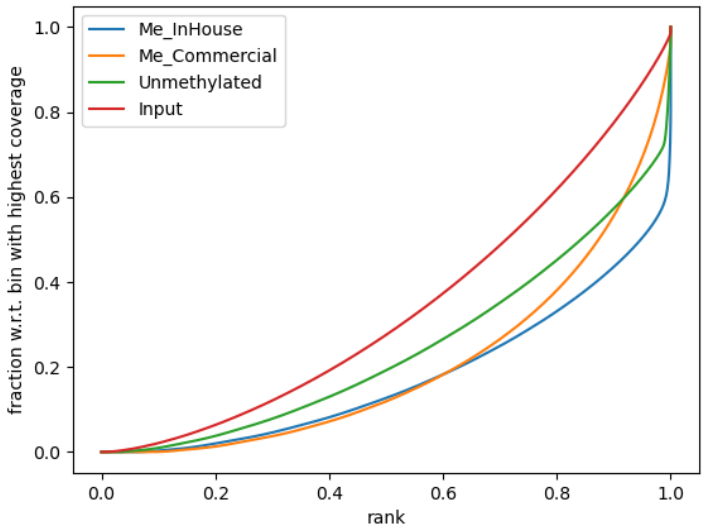


Figure 9. Fingerprint plot of the 4 capture methods. The x-axis represents the genomic bins ranked from lowest reads to highest. The y-axis represents a percentage scale of the cumulative fraction of total sequencing reads. The plot was used to determine how well the method captured its target DNA versus picking up background noise. The red Input line expected to lie very close to a perfect diagonal shows that reads are spread evenly across the genome. The green (unmethylated), yellow (commercial) and blue (in-house) lines indicate enrichment towards their targets confirming that MBD captured their targets better than the background fragments. M2 In-house (in blue) with the biggest 'elbow bent' shows that most methylated reads are concentrated in small genomic region which was intended for capturing.

DNA Methylation Profiles

Heatmaps generated across gene bodies showed strong signal enrichment at Transcription Start Sites (TSS) for methylated samples (M2 and M4), whereas the unmethylated control (M3) showed a distinct profile, and the Input remained flat. The enrichment profile across TSS and gene bodies validates the protein binding logic. M2 (‘In-house’ MBD) shows a characteristic "TSS Dip," where signal is low at the promoter (unmethylated) and high in the gene body (methylated). M3 (CXXC) shows a "TSS Peak," confirming the successful capture of unmethylated CpG islands typically found at promoters (Figure 10-13).

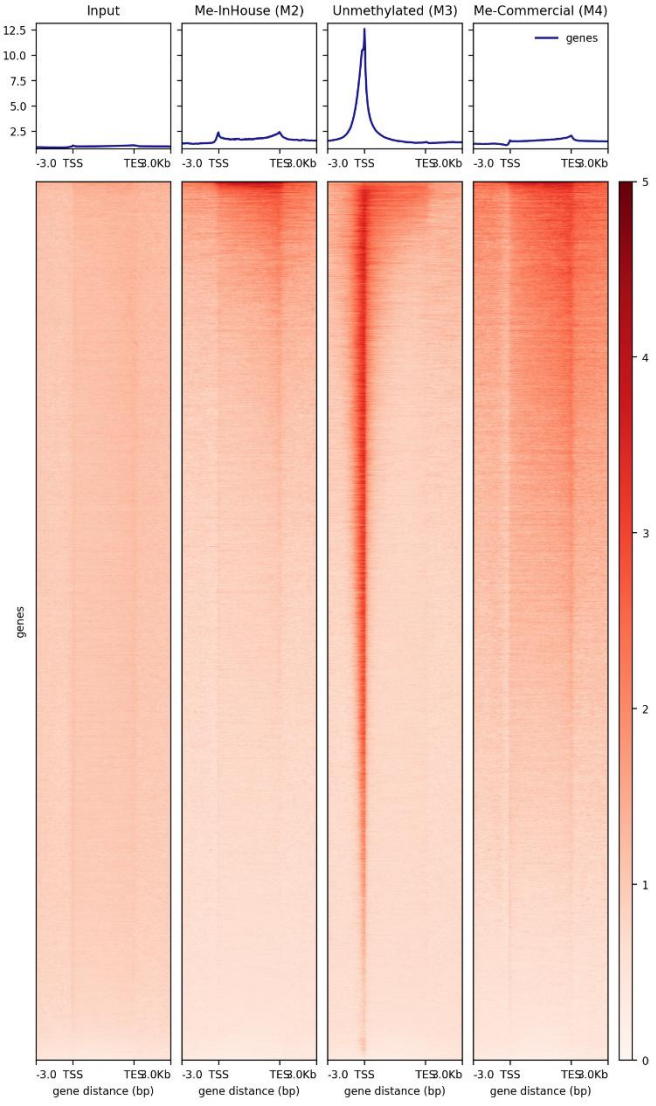


Figure 11. DNA methylation profile for Donor 1.

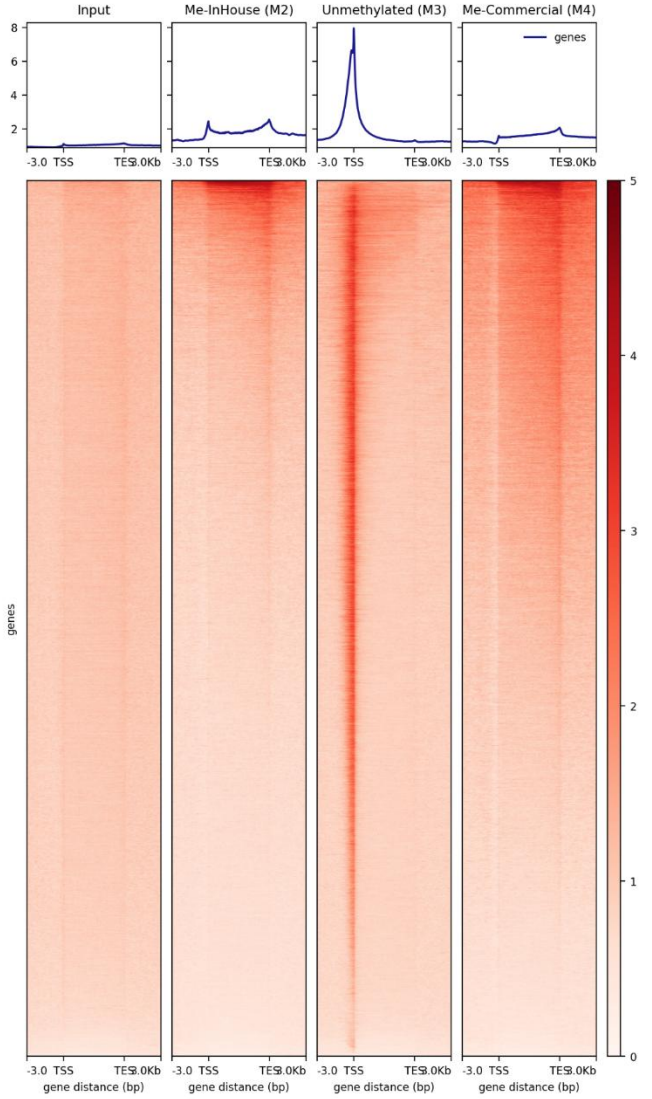


Figure 10. DNA methylation profile for Donor 2.

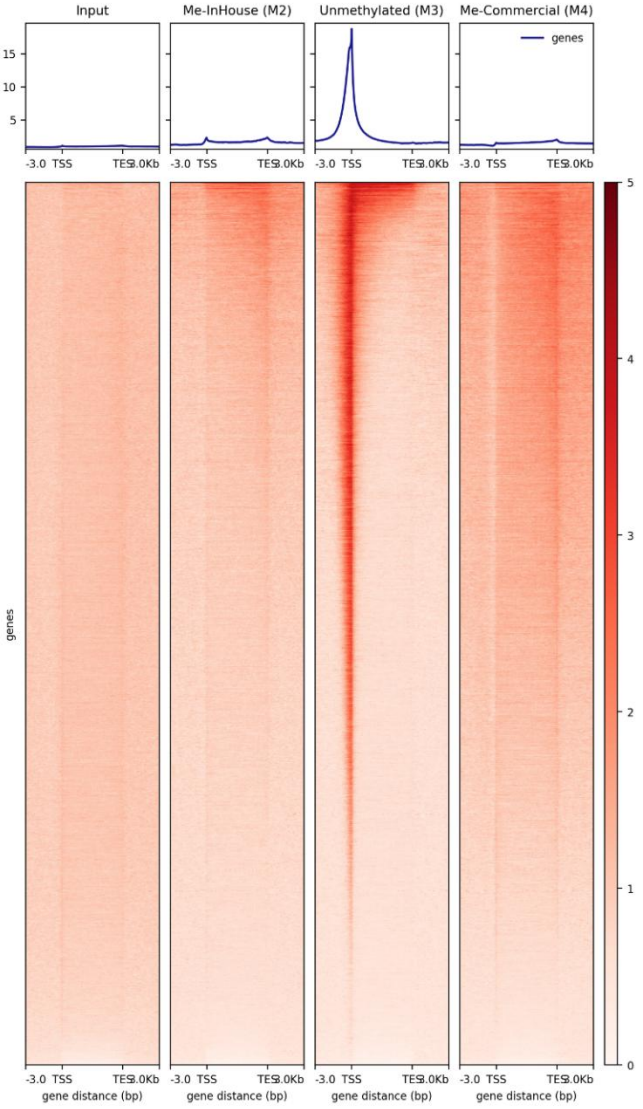


Figure 13. DNA methylation profile for Donor 3.

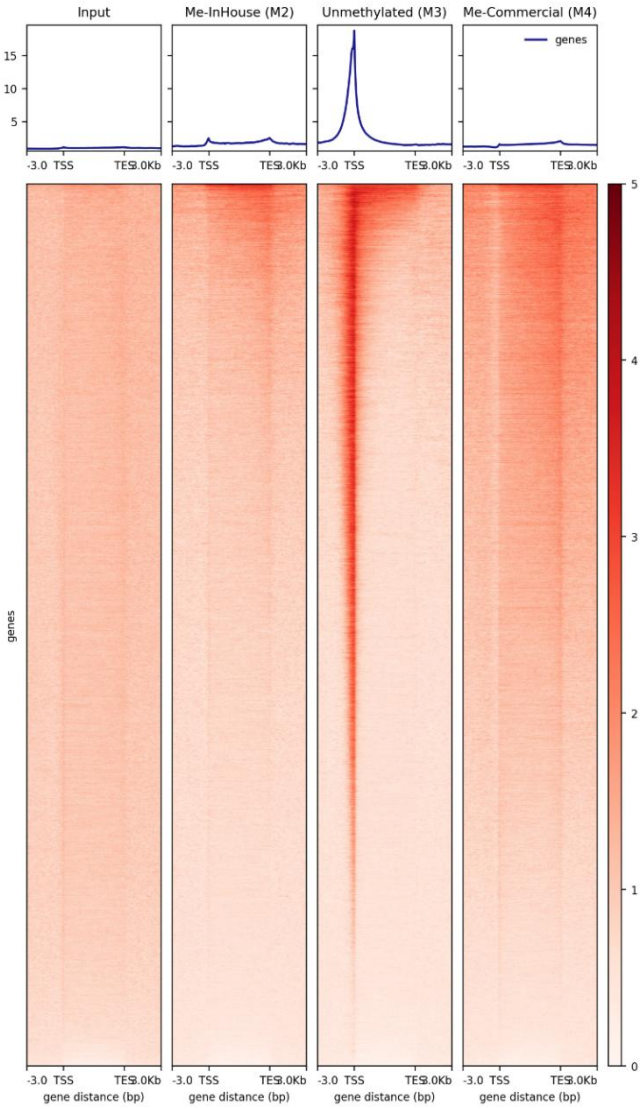


Figure 12. DNA methylation profile for Donor 4.

Feature distribution (Figure 14) illustrates the percentage of peaks located within specific genomic compartments relative to the hg19 reference genome, including promoters (defined as ± 3kb of the TSS), exons, introns, and distal intergenic regions. Both methods show significant enrichment in promoter-proximal regions, validating the affinity of MBD proteins for regulatory CpG islands. The In-House MBD (M2) exhibits a higher relative concentration of peaks within the immediate promoter (≤ 1kb), while the Commercial MBD (M4) demonstrates broader coverage across intronic and distal intergenic regions.

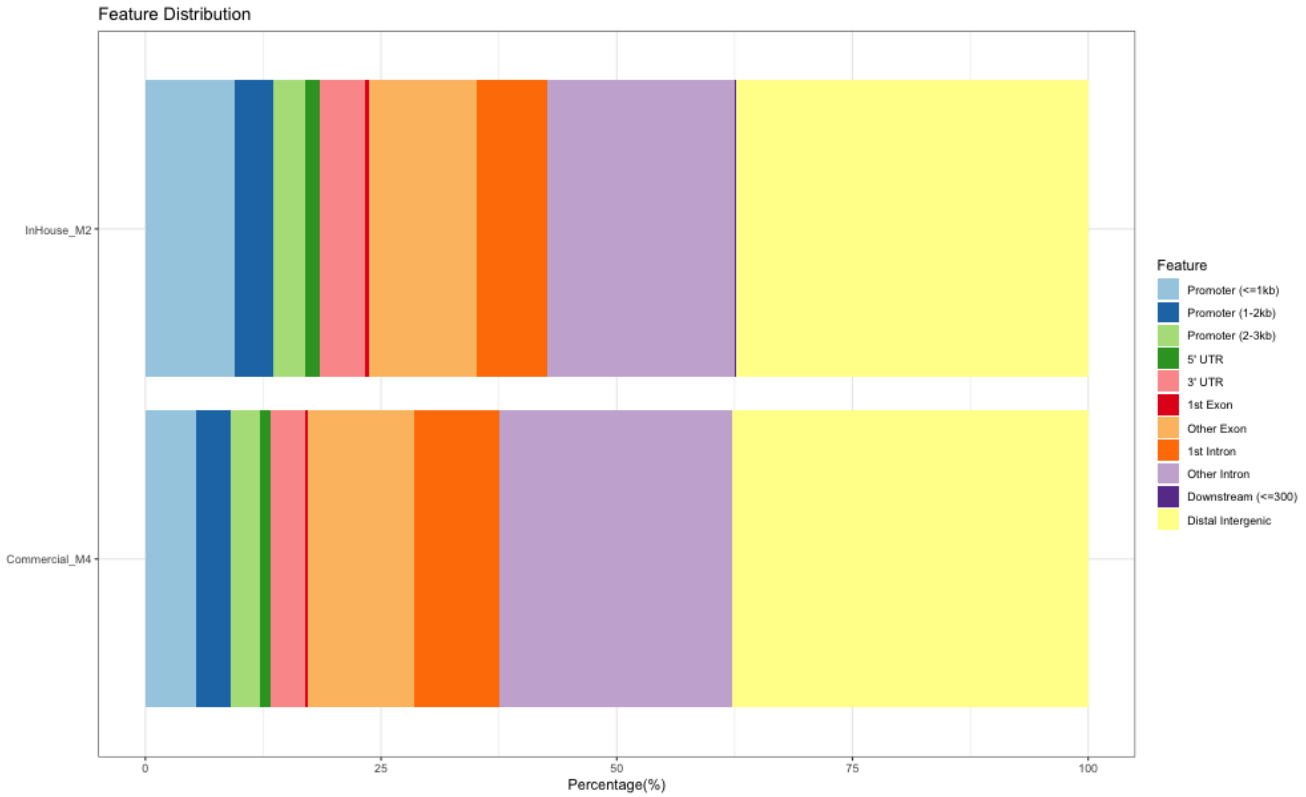


Figure 14. Genomic feature distribution of methylation enrichment peaks. Comparative annotation of enriched genomic regions identified by MACS3 for the In-House (M2) and Commercial (M4) MBD-capture methods.

Section 3: Cross-Platform Validation

Verification of Cell Identity

Annotation of unmethylated genes (M3) correctly identified active PBMC markers, including PTPRC (CD45), T-cell markers (CD3, CD4, CD8), and monocyte markers (CD14). Conversely, pluripotency markers (e.g., POU5F1, NANOG, SOX2) were identified as silenced/methylated in all MBD fractions, consistent with the differentiated state of PBMCs.

Comparative Analysis of Methylated Gene Sets (MBD vs. Array)

Total methylated genes identified when comparing the data from sequencing and microarray: In-House MBD (13,121), Commercial MBD (26,044), and 850K Array (20,026). To enable comparison between array-based and enrichment-based approaches, array probes with β-values > 0.6 were classified as methylated, approximating the high-methylation signal captured by MBD enrichment. Comparison of genes identified as "Methylated" (Beta > 0.6 for Array; Enriched for NGS) shows a core set of 6,888 genes common to all three methods. The In-House MBD method successfully captured 47% ((5433+6888)/26044x100) of the genes identified by the Commercial MBD kit (Figure 15). Additionally, the lists show that the identified genes (methylated) are involved in hematopoietic processes (e.g., pluripotency genes), confirming the silencing of irrelevant lineages in blood.

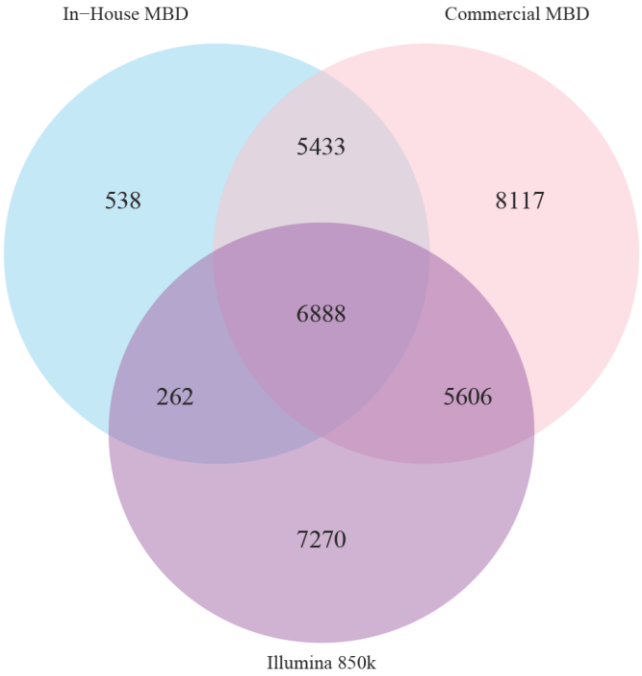


Figure 15. Venn diagram illustrating the overlap of methylated genes identified by In-house MBD, Commercial MBD, and the Illumina 850k array.

Discussion

The Utility of the MINUTE Pipeline

A significant achievement of this project was the implementation of the MINUTE pipeline. As this method has not been officially documented for this specific application before, the workflow was adapted from primary literature and optimized for PBMC samples. The results demonstrate that NGS barcoding and pooled processing provide a scalable alternative to traditional ChIP-seq, maintaining high signal-to-noise ratios while reducing technical variation between samples.

This project successfully re-engineered the MINUTE-ChIP pipeline, originally designed for histone marks, to accommodate MBD-seq data. By utilizing physical multiplexing, the pipeline reduces technical variability and batch effects at the source, offering a more robust alternative to purely mathematical correction methods like SVA.

Interpreting Genomic Profiles

The "TSS Dip" observed in M2 samples (Figures 10-13) is a critical indicator of methylome health. In PBMCs, most active genes (e.g., housekeeping genes) must remain unmethylated at the promoter to allow transcription. Our data shows that M2 accurately avoids these regions while capturing gene body methylation. Conversely, the high signal in M3 at the TSS proves that the CXXC protein successfully "fished" for the open, unmethylated promoters, providing a successful validation of the MBD results.

The genomic annotation analysis (Figure 14) reveals that both the In-House (M2) and Commercial (M4) pipelines successfully enriched for methylation at biologically relevant sites. Specifically, the high percentage of peaks localized to promoter regions (defined as ± 3kb from the TSS) confirms that the MBD-affinity capture successfully prioritized gene regulatory elements over non-functional background DNA. Interestingly, the In-House MBD (M2) method demonstrated superior specificity for proximal promoter regions (≤ 1kb from TSS) compared to the Commercial MBD (M4). While M4 captured a higher total number of peaks as previously seen in the Venn diagram (Figure 15), Figure 14 shows that a significant portion of that extra signal originated from distal intergenic and intronic regions. This indicates that while the commercial kit may have higher overall sensitivity, the In-House method offers a more 'focused' view of the core promoter methylome, which may be advantageous for studies focused strictly on transcriptional silencing.

Methodological Comparison: Array vs. NGS

The high overlap in the Venn Diagram (Figure 15) highlights the robustness of the In-House method.

Array Illumina 850k provides absolute quantification (Beta-values) but is limited to pre-defined probe locations and misses regions not covered by the 850,000 probes.

NGS MINUTE-seq provides a genome-wide "unbiased" view. It can identify methylation in repetitive elements or intergenic regions that the array ignores. However, data analysis is more complex, requiring normalization of read depths and handling of PCR duplicates. The high concordance (6,888 core genes) suggests that for clinical biomarker discovery, both methods are viable, but NGS offers a broader discovery potential.

The results indicate that while both MBD-seq and Illumina 850K arrays identify a significant core of identical methylated genes, MBD-seq provides regional enrichment data (peaks) rather than the site-specific direct measurements of the array. One observed limitation is the preference of MBD proteins for regions with high CpG density; lone methylated CpGs that are captured by arrays might be underrepresented in MBD-seq datasets.

The In-House MBD method (M2) was more conservative, identifying fewer genes than the commercial MBD (M4), yet showed high overlap with the 850K array results, suggesting high specificity. The enrichment of repressed genes in categories like transcriptional regulation further validates the biological accuracy of the captured methylation patterns.

Limitations and Technical Considerations

Each method showed specific limitations. As seen in Figure 7, the M4 fraction's proximity to Input suggests that "over-stringency" in MBD-seq can lead to low DNA complexity, where background noise eventually outweighs the specific signal.

While the array is highly reproducible, the "Gene List" generated from it is biased toward the UCSC RefGene annotations used for probe design.

MINUTE-ChIP resolution is limited by fragment size (~200-300bp), whereas the Array provides single-nucleotide resolution at the specific probe site.

The ability of the in-house MBD protein to bind CpG-rich regions is also dependent on the buffer concentration used, which still remains to be optimized.

The supplementary figures in the Appendix confirm that our adaptation of the MINUTE-ChIP pipeline worked effectively. The barcode representation (Figure 16) proves that the method for mixing multiple samples together (multiplexing) was successful. Each donor and condition was represented

fairly in the sequencing data, meaning no single sample overshadowed the others. This balance is further shown in the global read levels (Figure 17), where results remained consistent across all replicates.

Other technical figures support the very good quality of the data. Figure 21 shows that most sequencing quality scores (Phred scores) were very high (>30), meaning there was less than 1 in 1,000 chance that a base call is incorrect (99.9% accuracy). The GC content distribution (Figure 18) matches what we expected to see when capturing CpG-rich areas of the genome since this project uses MBD-seq to capture CpG sites, the peak GC content lies between 45-55%. Low GC content would mean that the capture picked up random, non-regulatory DNA. The insert size profile (Figure 19) shows that our DNA fragments were mostly between 150-200bp long which follows the typical insert size for the NextSeq 2000 P3 100-cycle kit used in this study. Finally, the chromosomal mapping (Figure 20) shows that the DNA reads were spread naturally across all chromosomes. Larger chromosomes like 1 and 2 contain more DNA and therefore we observe higher number of reads. We observe a peak in chromosome 19 which is expected as chr19 is known for being the more gene-rich, hence having a higher GC content as well. This ensured we have a complete, unbiased view of the entire methylome.

The Choice of the Method

Overall, the choice of the method will always depend on the scientific question and experimental needs. If one aims to determine the methylation status of a single gene, the Illumina 850k array is superior because it uses specific, fixed probes designed to target known genomic locations at promoters and CpG islands. The results give Beta-values which can be interpreted in a binary way, if the value is >0.6 then the region is hypermethylated (switched off), while <0.2 mean hypomethylated (switched on). Whereas MINUTE-ChIP shows relative enrichment of a region, where the signal depends on read depth and the density of neighbouring CpGs. On the other hand, if the goal is to identify a change in status like silencing of tumor suppressor genes or activation of oncogenes in cancer, MINUTE-ChIP offers better potential for exploration and marker discovery. For instance, in cancer if a previously unmethylated promoter would suddenly show a strong peak (indicating methylation), it would be easier to statistically detect it than a shift in an array's beta value. Additionally, MINUTE-ChIP is an unbiased method and can detect methylation changes anywhere in the genome, compared to the array that is limited by the number of predesigned probes.

Conclusion

This study successfully adapted the MINUTE-ChIP workflow for DNA methylation analysis in human PBMCs. We demonstrated that the In-House MBD method outperformed the commercial kit, offering better specificity for gene promoters without the loss of signal. Biological validation confirmed the method's accuracy, correctly distinguishing between active immune markers and silenced pluripotency genes. While the Illumina 850k array remains the gold standard for absolute quantification, this pipeline provides an unbiased alternative for genome-wide discovery. Ultimately, this protocol offers a scalable and cost-effective solution for high-throughput epigenetic research in precision medicine.

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- Maria Lerm, professor in Medical Microbiology at Linköping University who generously took me in for another exciting project,
- Sara Hosseini, Field Application Scientist at Epigenica, who advised on study design and facilitated collaboration between Epigenica and the research group.

Statement on usage of GAI

Google’s Gemini was used in this project to get an initial overview of the field, as a language tool to improve the vocabulary and grammar of the text, and as a troubleshooting tool for debugging the code in R.

Appendix

Supplementary figures

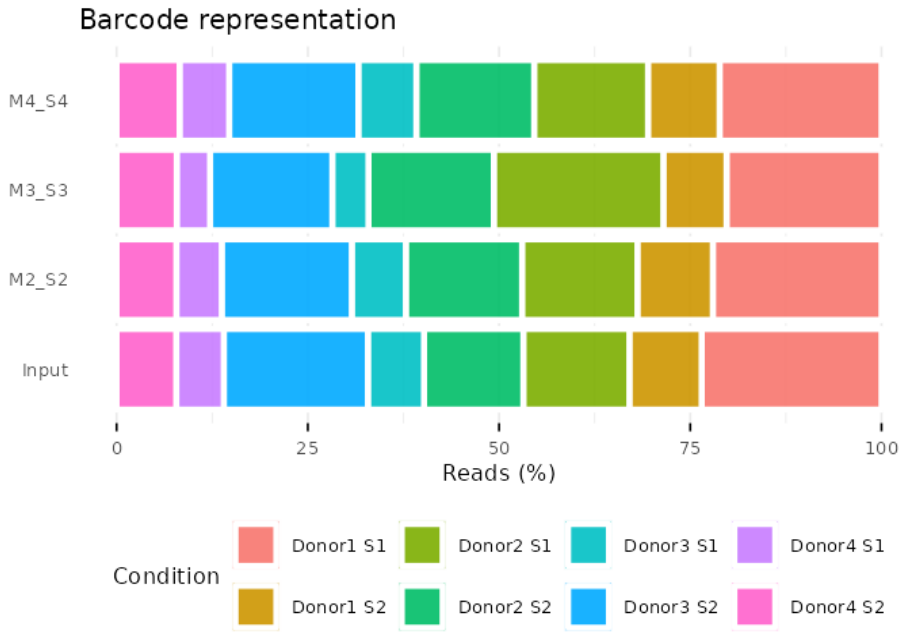


Figure 16. Normalized barcode representation (%).

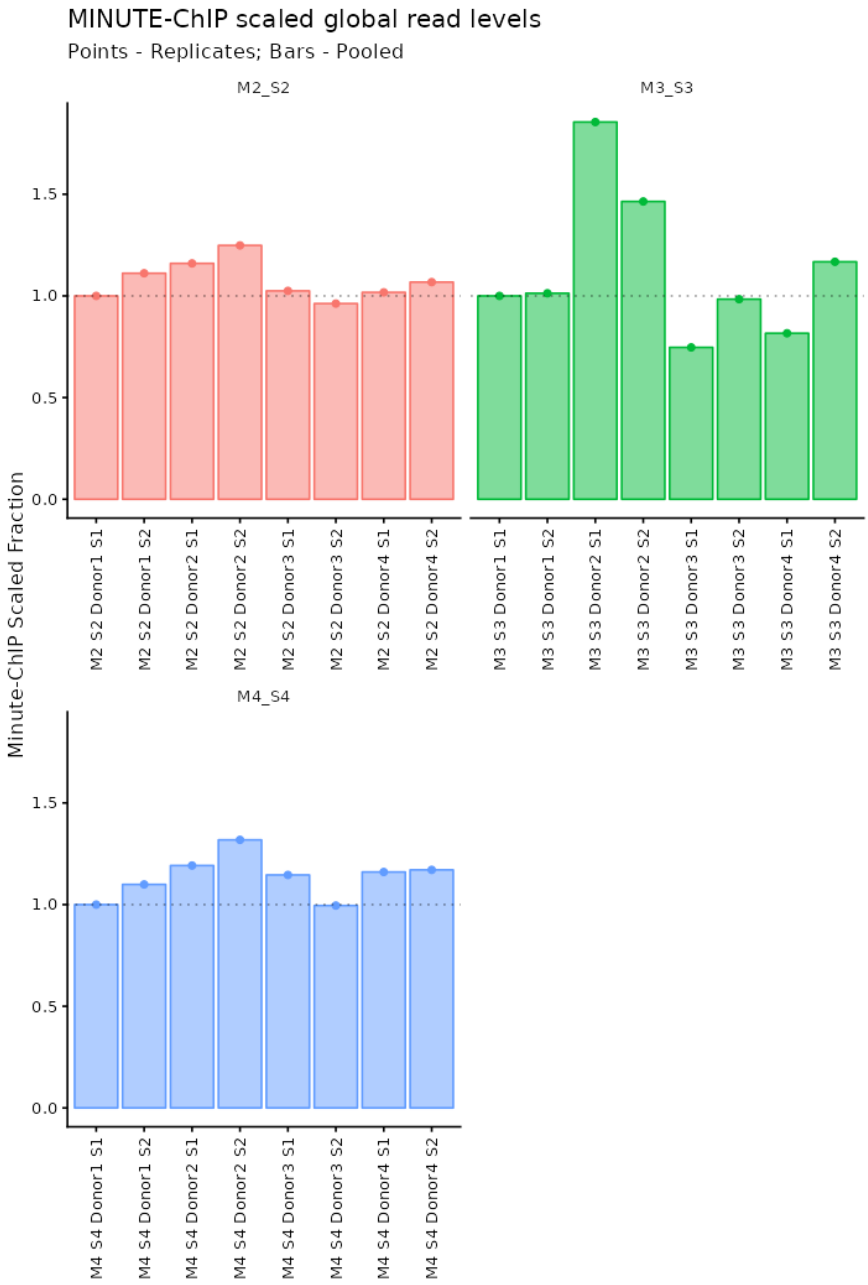


Figure 17. MINUTE-seq scaled global read levels.

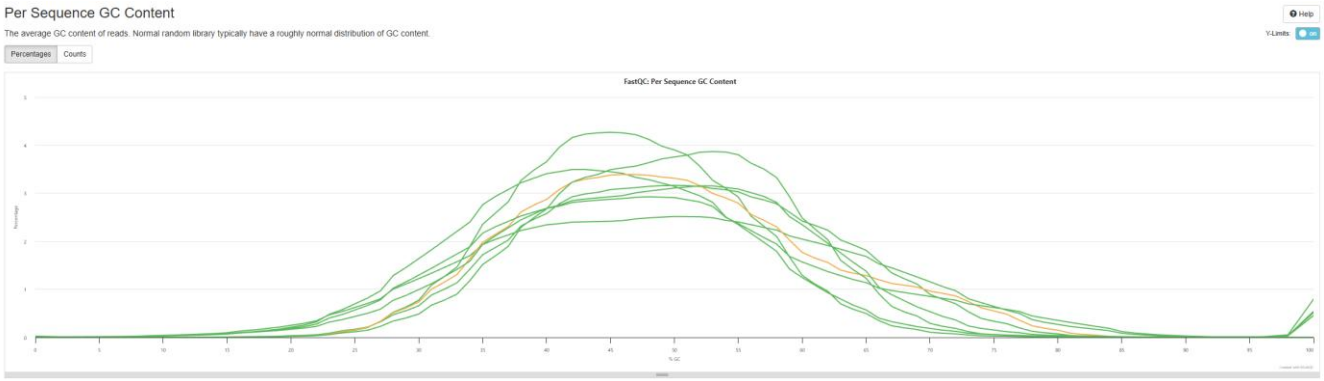


Figure 18. Average GC content (normalized, %).

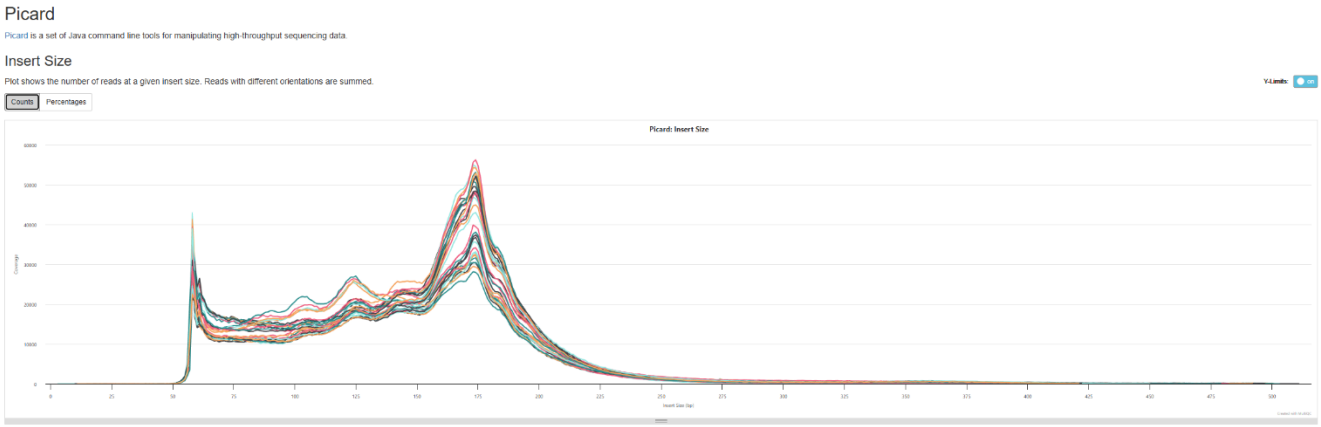


Figure 19. Picard plot showing number of reads at specific insert size.

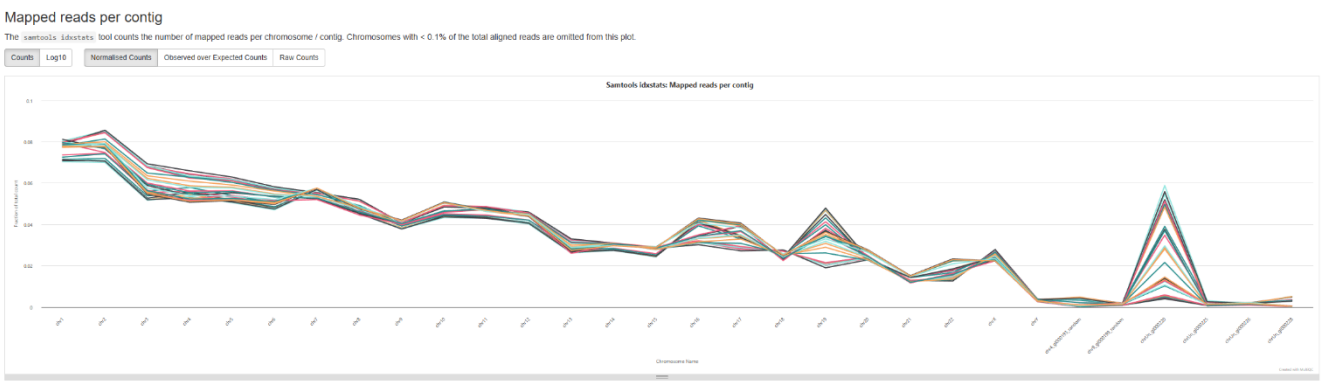


Figure 19. Number of mapped reads per chromosome using *samtools idxstats*.

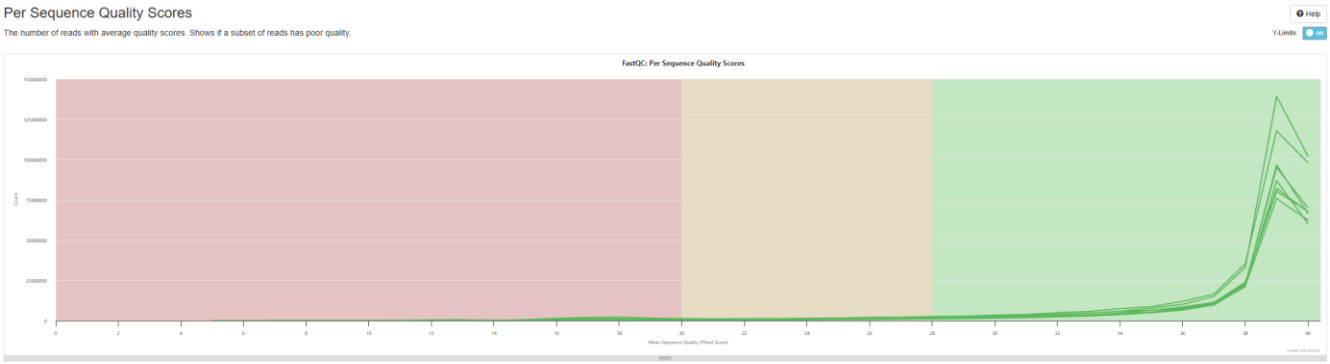


Figure 21. Number of reads with average quality scores.

References

Active Motif. (2022). MethylCollector™ MBD Capture Kit Manual (Version A1). Carlsbad, CA: Active Motif, Inc. Available at: www.activemotif.com

Cechova, M. (2020). Probably Correct: Rescuing Repeats with Short and Long Reads. *Genes*, *12*(1), 48. https://doi.org/10.3390/genes12010048

Cedar, H., & Bergman, Y. (2009). Linking DNA methylation and histone modification: Patterns and paradigms. *Nature Reviews Genetics*, *10*(5), 295–304. https://doi.org/10.1038/nrg2540

Fischer, N. (2020). Infection-induced epigenetic changes and their impact on the pathogenesis of diseases. *Seminars in Immunopathology*, *42*(2), 127–130. https://doi.org/10.1007/s00281-020-00793-1

Huoman, J., Sayyab, S., Apostolou, E., Karlsson, L., Porcile, L., Rizwan, M., … Lerm, M. (2022). Epigenetic rewiring of pathways related to odour perception in immune cells exposed to SARS-CoV-2 in vivo and in vitro. Epigenetics, 17(13), 1875–1891. https://doi.org/10.1080/15592294.2022.2089471

Kumar, B., Navarro, C., Yung, P. Y. K., Lyu, J., Salazar Mantero, A., Katsori, A.-M., Schwämmle, H., Martin, M., & Elsässer, S. J. (2025). Multiplexed chromatin immunoprecipitation sequencing for quantitative study of histone modifications and chromatin factors. Nature Protocols, 20(3), 779–809. https://doi.org/10.1038/s41596-024-01058-z

Leek, J. T., Scharpf, R. B., Bravo, H. C., Simcha, D., Langmead, B., Johnson, W. E., Geman, D., Baggerly, K., & Irizarry, R. A. (2010). Tackling the widespread and critical impact of batch effects in high-throughput data. Nature Reviews Genetics, 11(10), 733–739. https://doi.org/10.1038/nrg2825

Love, M. I., Huber, W., & Anders, S. (2014). Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. Genome Biology, 15(12), 550. https://doi.org/10.1186/s13059-014-0550-8

Mehrmohamadi, M., Sepehri, M. H., Nazer, N., & Norouzi, M. R. (2021). A Comparative Overview of Epigenomic Profiling Methods. Frontiers in Cell and Developmental Biology, 9, 714687. https://doi.org/10.3389/fcell.2021.714687

Pidsley, R., Zotenko, E., Peters, T. J., Lawrence, M. G., Risbridger, G. P., Molloy, P., Van Djik, S., Muhlhausler, B., Stirzaker, C., & Clark, S. J. (2016). Critical evaluation of the Illumina

MethylationEPIC BeadChip microarray for whole-genome DNA methylation profiling. Genome Biology, 17(1), 208. https://doi.org/10.1186/s13059-016-1066-1

Qvick, A., Adolfsson, E., Tornéus, L., Lindqvist, C. M., Carlsson, J., Stenmark, B., Karlsson, C., & Helenius, G. (2025). Pan-Cancer Detection Through DNA Methylation Profiling Using Enzymatic Conversion Library Preparation with Targeted Sequencing. International Journal of Molecular Sciences, 26(20), 10165. https://doi.org/10.3390/ijms262010165

Turner, B. M. (2009). Epigenetic responses to environmental change and their evolutionary implications. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *364*(1534), 3403–3418. https://doi.org/10.1098/rstb.2009.0125

Whalley, C., Payne, K., Domingo, E., Blake, A., Richman, S., Brooks, J., Batis, N., Spruce, R., S-CORT Consortium, Mehanna, H., Nankivell, P., & Beggs, A. D. (2021). Ultra-Low DNA Input into Whole Genome Methylation Assays and Detection of Oncogenic Methylation and Copy Number Variants in Circulating Tumour DNA. *Epigenomes*, *5*(1), 6. https://doi.org/10.3390/epigenomes5010006

Yu, X., Zhao, H., Wang, R., Chen, Y., Ouyang, X., Li, W., Sun, Y., & Peng, A. (2024). Cancer epigenetics: From laboratory studies and clinical trials to precision medicine. *Cell Death Discovery*, *10*(1), 28. https://doi.org/10.1038/s41420-024-01803-z

Zhang, Q., & Cao, X. (2019). Epigenetic regulation of the innate immune response to infection. *Nature Reviews Immunology*, *19*(7), 417–432. https://doi.org/10.1038/s41577-019-0151-6